The results are plotted in the parameter mapping shown in Figure 5. The solid lines represent the boundaries in ($N_R - a\kappa$) space for the existence of primary minima for various values of the parameter N_{12}. For values of N_R greater than those given by a particular solid curve, no primary minimum exists for that value of N_{12}. The value of N_R at which the primary minimum disappears decreases with increasing N_{12}. The dashed line provides the boundary in ($N_R - a\kappa$) space for the existence of a secondary minimum. For values of N_R less than those given by the dashed curves, no secondary minimum exists. The location of this line was found to be fairly insensitive to the value of N_{12} chosen. Curves for various N_{12} superimposed onto the single dashed curve shown except near the intersection points with the solid lines where the curves merged.

For each value of N_{12}, the lines shown in Figure 5 map ($N_R - a\kappa$) space into three regions. Above a given solid line only secondary minima exist for that value of N_{12}. Below the dashed line, the interparticle potential curves exhibit only a primary minimum. The region between the dashed and a solid curve corresponds to a case with both minima being present.

Figure 5 can be used to predict stability states of colloid dispersions. For example, should a potential with both a primary and a secondary minimum be required, the constraints on N_R, N_{12}, and $a\kappa$ can be determined by inspection.

To illustrate the effects of incorporating Born repulsive forces, interparticle potentials were calculated with and without the Born contribution. Results are shown in Figure 6. For the case of weak electrostatic repulsion, neglect of the Born term results in an interparticle potential that is unbounded and negative upon particle contact. Inclusion of the Born term restricts the primary minimum to finite depth. When there is a significant degree of electrostatic repulsion, as is shown in Figure 6, Born forces may completely eliminate the primary minimum. The weak secondary minimum in this case is present outside of the range of this graph.

It is also interesting to note that although Born forces are short ranged, they can alter the shape of the interparticle potential diagram in the vicinity of the classical interparticle maximum. As shown in Figure 6, the maximum occurs at a dimensionless interparticle separation of ~0.0045, which corresponds to a separation of roughly 20 Å for 1-μm-diameter particles. Similar effects are not observed when the Born repulsion is modeled as a hard-sphere interaction.

We have demonstrated that inclusion of a noninfinitely steep Born repulsion between macroscopic particles may result in the elimination of the primary minimum in the interaction energy curve under realistic conditions. This result was developed by assuming pairwise additivity of the interatomic potentials and additivity of electrostatic, Born, and van der Waals potentials. Although the question of the validity of the pairwise additivity assumption has been explored for van der Waals attractive potentials, there has apparently been no work done on the additivity of repulsive forces. There is no reason to expect the repulsive forces to follow additivity rules any more than the attractive forces do. However, the formulas presented do have better qualitative characteristics of the repulsive interactions between macroscopic particles than either the hard-sphere, square-step, or the R^{-12} potentials that have been used in the past. Although it may be unreasonable to expect that the electron states of the atoms remain unperturbed when particles are in each others' vicinity, we believe our formulas do offer a practical alternative description of the Born forces within colloidal systems for those cases wherein detailed knowledge of close-range forces is not needed. In addition, it is apparent that Born forces should not be ignored for realistic modeling of colloidal interactions.

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Mutual Diffusion Coefficients and Ionic Transport Coefficients l_{ij} of $MgCl_2$–H_2O at 25 °C

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The volume-fixed mutual diffusion coefficients of aqueous $MgCl_2$ have been accurately measured at 25 °C from dilute solutions to near saturation, by free-diffusion Rayleigh interferometry. Density data for $MgCl_2$–H_2O have been measured with pycnometers. The generalized ionic transport coefficients of irreversible thermodynamics, l_{ij}, have been calculated from our diffusion data and critically reviewed data for thermodynamic and other transport properties. These l_{ij} values are compared with those of other 2–1 chlorides. The diffusion data, l_{ij} values, and results reported elsewhere form part of the data base necessary for development of approximation procedures for binary and ternary solution transport properties. Mutual diffusion data are also reported for several sucrose–H_2O solutions. Realignment of the Beckman-Spinco Model-H electrophoresis apparatus to optimize Rayleigh interferometry is described in the supplementary material.

I. Introduction

The substance $MgCl_2$ is the second largest electrolyte constituent of seawater[4] and is a major component of many deep subsurface brines. Interest in desalination resulted in much data on seawater, its constituent salts, and their mixtures, but most of it has been for equilibrium properties. Consequently, diffusion coefficient data to near saturation for these solutions are essential both for practical applications and for a complete understanding of transport in these brine systems. Such data are already available for NaCl.[5]

Harned and Polestra[6] used the restricted diffusion conductance method[7] to measure diffusion coefficients of $MgCl_2$–H_2O from 0.001 to 0.004 mol·dm^{-3} at 25 °C. Richardson et al.[8] reported

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diaphragm cell data from 0.01 to 0.55 mol·dm^{-3} for this salt (the solubility is about 4.99 mol·dm^{-3}). Unfortunately, diaphragm cell experiments yield integral diffusion coefficients which must be differentiated to yield the desired differential diffusion coefficient D_v. Such differentiations can yield 20–50% errors in D_v for unfavorable cases.[9,10] Optical interferometry measurements are thus warranted for $MgCl_2$–H_2O since they yield much more accurate values of D_v.

In this report Rayleigh interferometry measurements of the differential diffusion coefficients of $MgCl_2$–H_2O are given at 25 °C up to near saturation (4.9 M). They complement measurements for other alkaline-earth chlorides[5,11,12] and also complete the measurements of the various isothermal transport properties of aqueous $MgCl_2$ at 25 °C. Diffusion coefficients D_v, cation transference numbers t_1, and equivalent conductances Λ are critically reviewed here and used to calculate the generalized ionic transport coefficients l_{ij} of irreversible thermodynamics.

The l_{ij} coefficients (Onsager coefficients) are based on a general linear description of all types of transport processes. Therefore, once given the l_{ij} as functions of composition, temperature, and pressure for an electrolyte system, any linear transport process no matter how complex can be accurately described. Because of the generality of the theory, the behavior of the l_{ij} should give different and perhaps better insights into electrolyte transport.[13–16] Consequently, $MgCl_2$ l_{ij} coefficients are compared with those of other 2–1 chlorides. In addition, Altenberger and Friedman[17] have noted that the l_{ij} coefficients are a more sensitive test of electrolyte models than is the conductance. Moreover, these coefficients for binary systems can be used to estimate transport properties in mixtures at finite concentrations.[14,15,18] We intend to use the l_{ij} reported here in the analysis of the NaCl–$MgCl_2$–H_2O system, whose diffusion coefficients are being investigated in this laboratory.

II. Experimental Section

The experimental diffusion and density measurements were performed at 25.00 ± 0.005 °C (IPTS-68) over a period of several years using five separately prepared and analyzed $MgCl_2$ stock solutions. The molecular weights used were 95.211 g·mol^{-1} for $MgCl_2$, 74.556 g·mol^{-1} for KCl, and 342.299 g·mol^{-1} for sucrose.

$MgCl_2$ samples for preparation of stock solutions were purified by saturating solutions with $MgCl_2$·$6H_2O$ at 50 °C, followed by cooling to 0 °C and removal of crystals. Stock 1 was prepared from a mixed lot of "Baker Analyzed" reagent grade and Mallinckrodt analytical reagent $MgCl_2$·$6H_2O$. Stocks 2, 3, and 5 were prepared from Mallinckrodt analytical reagent, and stock 4 was prepared from "Baker Analyzed" $MgCl_2$·$6H_2O$. Lower concentrations were prepared by mass dilution; higher concentrations were obtained by removal of solvent in a desiccator. The water was first deionized and then distilled.

Results for the concentration and trace impurity analyses of stocks 1 and 2 have been reported elsewhere (same stock numbering system used).[19] These impurities were only 0.003% by weight. These stock solutions, and stocks 3 and 5, were analyzed by the sulfate method for magnesium and by $AgNO_3$ titration for chloride.[19] The concentrations of these stocks are known to 0.05% or better.

The concentration of stock 4 (chronologically the first one prepared) was obtained by coulometric analysis for the chloride.

TABLE I: Densities of Aqueous $MgCl_2$ Solutions at 25 °C

m, mol·kg^{-1}	c, mol·dm^{-3}	d, g·cm^{-3}
0.30313	0.30061	1.02032
0.50643	0.50023	1.03538
1.0242	1.0004	1.07203
1.5552	1.5001	1.10742
2.1583	2.0504	1.14523
2.7828	2.6004	1.18204
3.2528	3.0009	1.20829
3.6128	3.3004	1.22778
4.1075	3.7013	1.25352
4.4874	4.0014	1.27267
4.7443	4.2004	1.28527
5.0075	4.4009	1.29789
5.2459	4.5798	1.30907
5.5115	4.7758	1.32122
5.6307	4.8625	1.32653
5.7807	4.9705	1.33310
3.5011	3.2079	1.22171[a]
3.5016	3.2084	1.22174[a]
5.5667	4.8162	1.32374[a]

[a] These three samples were measured with stock 5; the other densities are for stock 2.

This analysis is believed to be much less accurate than those used for the other stock solutions.

The sucrose solutions were prepared by mass from a National Bureau of Standards "standard sample".

The densities of samples from $MgCl_2$ stocks 2 and 5 were measured in duplicate with ~31-cm^3 single-stem pycnometers. The mean values of the densities are listed in Table I and are precise and concordant to at least 0.000 03 g·cm^{-3}. However, uncertainty in the density due to stock solution concentration uncertainty could be as large as 0.000 13 g·cm^{-3} at the highest concentration but is negligible at low concentrations.

Diffusion measurements were performed using a Beckman-Spinco Model-H electrophoresis apparatus operated as a free diffusion Rayleigh interferometer. The earliest series of $MgCl_2$–H_2O measurements (dilutions from stock 4) was performed using the instrument in its standard alignment mode. This alignment is not optimal for Rayleigh interferometry. Consequently, the optical constant, $\beta = 4.180 \times 10^{-5}$ cm^2·min·mm^{-2}·s^{-1}, was obtained by calibration with KCl–H_2O data using the procedure of Albright and Miller with the same KCl sample.[9] No base-line corrections were applied to these KCl–H_2O and $MgCl_2$–H_2O measurements. The precision of β, and the corresponding $MgCl_2$–H_2O diffusion data, is about ±0.25% for this series of measurements.

After these initial measurements were complete, the optical arrangement of the diffusiometer was realigned to optimize the Rayleigh mode. This allowed a more accurate absolute value of β to be determined. The new β is 4.1864×10^{-5} cm^2·min·mm^{-2}·s^{-1} and was obtained with a precision of about 0.01% by photographing a ruled scale in the center-of-cell position.[5]

The remaining $MgCl_2$–H_2O and all sucrose–H_2O experiments were performed with this new alignment, and base-line corrections were made. All experiments but one were performed using a modified Beckman diffusion cell. The accuracy of these binary solution D_v measurements is 0.1–0.2% except for the lowest concentration point, which is uncertain by 0.5–0.8% owing to the small number of fringes.

Many Beckman-Spinco Model-H electrophoresis apparatus are still available for potential use as Rayleigh diffusiometers. As a guide to future workers, the realignment procedure, described briefly elsewhere,[5,20] is given in more detail in the supplementary material.

The method of reading photographic plates and the calculation procedure for binary solutions were the same as given by Albright and Miller,[9] except for the application of base-line corrections to most of the measurements. For dilute electrolyte solutions the

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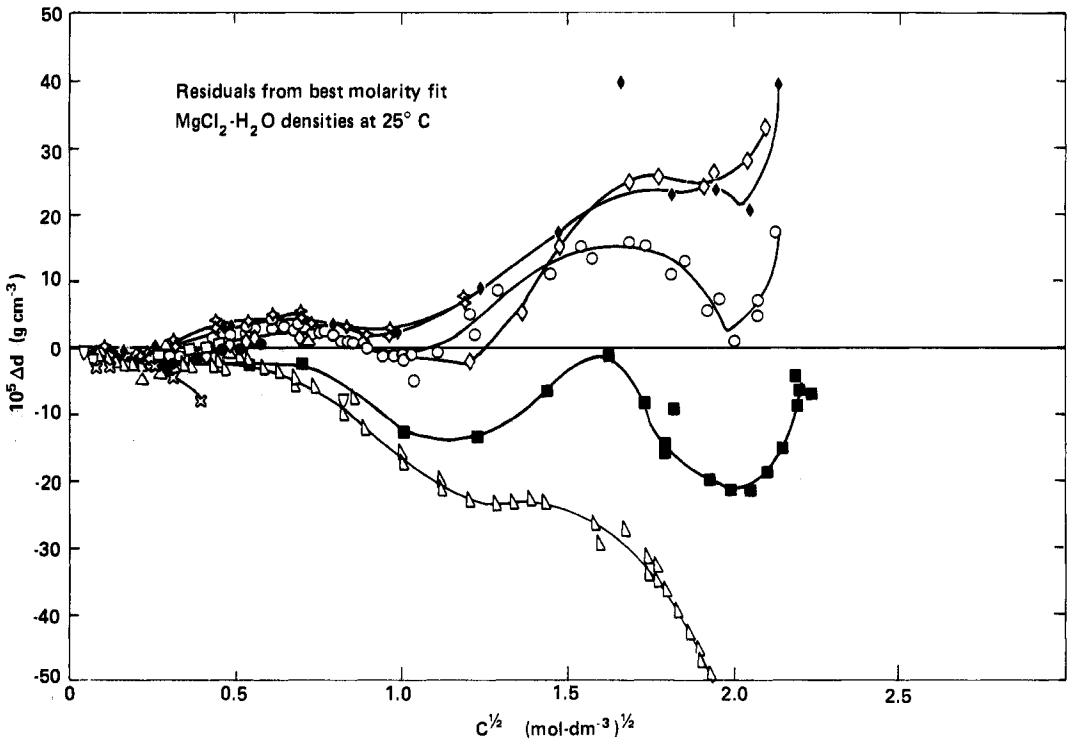


Figure 1. Differences between experimental densities and the least-squares molarity fit for aqueous $MgCl_2$ at 25 °C: ■, this research; ●, ref 32; ◺, ref 29; ○, ref 26; ◇, ref 34; ◆, ref 33; ✧, ref 35; ×, ref 25; △, ref 23; □, ref 28; ▽, ref 27.

procedure of Miller et al.,[21] briefly discussed elsewhere,[20] was used to correct for concentration dependences of the diffusion coefficients and refractive indexes.

The refractive index values are much more precise for measurements after the optical realignment than for the initial series. Consequently, only values measured with the realigned instrument (Table III, series A, D, E, and F) were used to compute the low-concentration corrections.

III. Results

The $MgCl_2$–H_2O dilutions were prepared by mass, and density values are required to interconvert the molarity and molality concentration scales. Examination of available density data[22-36] indicated surprisingly large differences among some of these studies. Consequently, density measurements were performed for this system from 0.303 to 5.781 mol·kg^{-1}, and the results are presented in Table I. In this paper, mol·kg^{-1} denotes molality units mol·(kg H_2O)$^{-1}$.

TABLE II: Polynomial Coefficients for $MgCl_2$ Density Equations[a]

term	m, mol·kg^{-1}	c, mol·dm^{-3}
A_2	0.082 241 19	0.082 947 10
A_3	−0.015 919 29	−0.018 765 29
A_4	0.019 551 86	0.026 802 78
A_5	−0.018 689 97	−0.024 313 55
A_6	0.007 289 804	0.009 888 612
A_7	−0.001 029 203	−0.001 481 259
σ	0.000 099	0.000 083
max concn	5.7807	4.9705

[a] 225 points were given nonzero weights in the least-squares fits.

Available density data for $MgCl_2$–H_2O were fitted by the equation

$$d = d_0 + \sum_{i=2}^{n} A_i X^{i/2} \quad (1)$$

where d_0 = 0.997 045 g·cm^{-3} is the density of pure H_2O[37] and X is the molarity c in mol·dm^{-3} or the molality m in mol·kg^{-1}. Literature density values, when necessary, were converted to g·cm^{-3} and to our molecular weight of $MgCl_2$.

The data in Table I and the results of Shedlovsky and Brown,[23] Lee,[28] Dunn,[27] and Lo Surdo et al.[26] were given unit weights in the least-squares fits. Other density data were weighted 0.5,[25,32-35] 0.2,[29] or zero.[22,24,30,31,36] The lower weight assigned to Wirth and Bangert[29] was due to a large amount of data being based on a single chemical analysis. The parameters for eq 1 are given in Table II.

Five other sets of density data are in reasonably good agreement with ours, ±0.05%.[26-29,33,38] (Figure 1). High-concentration data sets generally show parallel trends but are displaced from one another. Stock concentration uncertainty and possible impurities in other studies may explain this. Other density data[24,30,31,36] (not shown in Figure 1) show gross systematic deviations from the more accurate results. Lo Surdo et al.'s data extend only to 5.27 mol·kg^{-1}, not to saturation (5.81 mol·kg^{-1}) as they claim.

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TABLE III: Diffusion Coefficients of Aqueous $MgCl_2$ Solutions at 25 °C

$\bar{c}$, mol·dm^{-3}	Δc, mol·dm^{-3}	$m(\bar{c})$, mol·kg^{-1}	J	$10^5 D_v$, cm^2·s^{-1}	$10^5 M$, cm^2·s^{-1}	stock and alignment[a-c]
0.003644	0.007288	0.003655	16.57	(1.142_5)	(1.256)	A, D
0.004828	0.009656	0.004843	22.00	1.135_3	1.260	A, F
0.005976	0.011952	0.005994	26.47	1.132_7	1.267	B, G
0.010961	0.021922	0.010995	49.59	1.104_9	1.263	A,
0.022436	0.044872	0.022510	100.96	1.081_6	1.263	B, G
0.029740	0.059480	0.029841	133.76	1.071_4	1.257	B, G
0.049797	0.039810	0.049983	88.19	1.056_6	1.244	A, F
0.075673	0.042317	0.075989	92.24	1.051_6	1.230	B, G
0.099526	0.039738	0.099983	87.11	1.043_5	1.209	A, F
0.15677	0.046738	0.15765	101.76	1.039_1	1.171	B, G
0.19867	0.039586	0.19994	85.44	1.038_3	1.144	A, F
0.26995	0.061267	0.27204	132.00	1.044_6	1.104	B, G
0.29751	0.03944	0.29998	84.06	1.042_8	1.084	A, F
0.39590	0.04131	0.39995	87.10	1.049_8	1.027	A, F
0.40406	0.05135	0.40826	106.88	1.049_8	1.022	B, G
0.49987	0.03905	0.50605	83.32	1.063_4	0.974	A, C, E
0.59286	0.04747	0.60136	97.03	1.069_2	0.922	B, G
0.78667	0.04641	0.80131	93.41	1.083_2	0.826	B, G
0.80165	0.04942	0.81685	100.57	1.086_5	0.821	B, G
0.97161	0.04896	0.99391	98.09	1.096_7	0.747	A, D
1.1730	0.04299	1.2058	84.83	1.114_7	0.673	B, G
1.4489	0.04992	1.4999	96.25	1.120_9	0.579	A, D
1.4543	0.05000	1.5057	95.18	1.121_6	0.578	B, G
1.7160	0.06081	1.7892	115.30[d]	1.124_6[d]	0.504	B, G
1.9082	0.04888	2.0003	89.49	1.120_8	0.456	A, D
2.3543	0.04456	2.5004	81.84	1.094_9	0.363	A, D
2.6102	0.06000	2.7941	106.49	1.071_6	0.321	B, G
2.8679	0.04112	3.0950	72.90	1.026_4	0.280	A, D
3.1797	0.05424	3.4664	94.88	0.966_3	0.238	A, D
3.6277	0.05052	4.0147	86.80	0.860_1	0.185	A, D
4.0113	0.05145	4.4991	86.92	0.760_9	0.147	A, D
4.3969	0.05257	5.0010	87.52	0.655_8	0.112	A, D
4.5553	0.05038	5.2121	83.42	0.611_3	0.099_3	A, E
4.6561	0.05899	5.3480	97.05	0.585_6	0.091_8	A, D
4.9229	0.05832	5.7140	93.27	0.515_0	0.073_2	A, D

[a] Series A measurements done with realigned optics and base-line corrections. Series B measurements done with original optical alignment and no base-line corrections. For series C (one experiment) Ta and glass cell used. All other experiments were in a Beckman cell. [b] D denotes stock 1, E denotes stock 2, F denotes stock 3, and G denotes stock 4. [c] The first nine values of D_v calculated directly from the fringe patterns before correction were respectively 1.145_0, 1.138_1, 1.135_8, 1.108_8, 1.086_4, 1.076_3, 1.057_2, 1.051_8, and 1.043_6 cm^2·s^{-1}. The corrections were based on the following equations to represent D_v and refractive index: $D_v = D°(1 + \alpha_1 c^{1/2} + ... + \alpha_4 c^2)$ and $n = n° + c(\rho_0 + \rho_1 c^{1/2} + ... + \rho_4 c^2)$, with α_1–α_4 as −1.670, 6.911, −13.34, 9.687, respectively, and ρ_0–ρ_4 as 0.025 34, −0.005 77, 0.011 62, −0.015 41, 0.007 51, respectively. [d] These photographic plates were reread, and $J = 115.29$ and $10^5 D_v = 1.125_1$ cm^2·s^{-1} were obtained.

The experimental $MgCl_2$–H_2O diffusion data are listed in Table III. D_v is the volume-fixed mutual diffusion coefficient in units of cm^2·s^{-1}, $\bar{c}$ is the average molar concentration in mol·dm^{-3} of the pair of solutions used in the diffusion experiment, Δc is the concentration difference across the initial diffusion boundary in mol·dm^{-3}, and J is the total number of Rayleigh fringes. The thermodynamic diffusion coefficient M is based on the driving force of irreversible thermodynamics, namely the chemical potential gradient. M is given by

$$M = D_v/(\mathrm{d}(m\Phi)/\mathrm{d}m) = D_v/(1 + m\ \mathrm{d}\ \ln\ \gamma/\mathrm{d}m) \quad (2)$$

Molal osmotic coefficients, Φ, and derivatives were calculated from the equation given elsewhere.[19] Here γ is the mean molal activity coefficient.

The two series of measurements reported in Table III are in complete agreement with each other and also connect up fairly well with Harned and Polestra's low-concentration conductometric results.[6] Six optical points from our earliest (less precise) measurements, which are somewhat high (0.3–0.7%), were given less weight in the smoothings described in section IV.

Values of D_v for $MgCl_2$–H_2O, obtained by differentiation of the Richardson et al. diaphragm cell integral diffusion coefficients,[8] are about 4% low at 0.1 mol·dm^{-3} and 12% high at 0.5 mol·dm^{-3}. The systematic errors for these diaphragm cell results are similar to those observed for $CaCl_2$–H_2O.[39,40] This suggests that inaccuracies in the differentiation of diaphragm cell data are the major source of error for both systems.

TABLE IV: Diffusion Coefficients of Aqueous Sucrose Solutions at 25 °C

$\bar{c}$, mol·dm^{-3}	Δc, mol·dm^{-3}	$m(\bar{c})$, mol·kg^{-1}	J	$10^5 D_v$, cm^2·s^{-1}
0.005 132	0.010 264	0.005 153	46.28	0.523_4
0.010 897	0.021 794	0.010 954	97.91	0.520_8
0.011 096	0.022 192	0.011 155	100.22	0.520_1
0.025 296	0.023 996	0.025 507	107.82	0.516_9
0.025 296	0.023 996	0.025 507	107.70	0.515_8

Figure 2 shows the D_v of Table III and those of Harned and Polestra,[6] as well as smoothed values of M. The features of both curves are typical of strong 2–1 electrolytes.

The diffusion coefficients of several dilute sucrose solutions, obtained over the past decade, are reported in Table IV. These diffusion coefficients are in good agreement with other optical values,[41-51] with the maximum difference from Chatterjee's Jamin interferometry results[48] being 0.5%. Values from Ingelstam–

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(51) Gosting, L. J.; Kim, H.; Loewenstein, M. A.; Reinfelds, G.; Revzin, A. *Rev. Sci. Instrum.* **1973**, *44*, 1602.

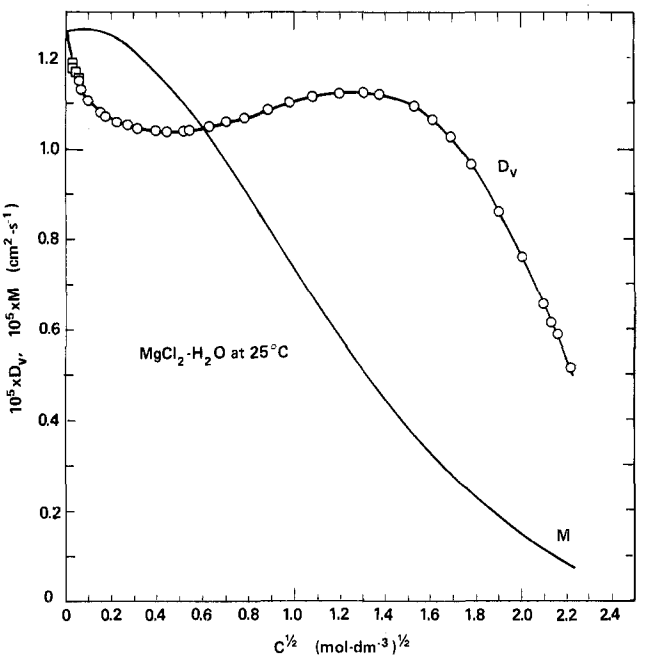


Figure 2. Mutual diffusion coefficients, D_v, and thermodynamic diffusion coefficients, M, of aqueous $MgCl_2$ at 25 °C as a function of the square root of the molarity: O, this research; □, ref 6.

Bryngdahl shearing diffusiometry[52] show larger differences but are less precise measurements. The densities of sucrose–H_2O were taken from Timmermans.[53]

IV. Critical Review of $MgCl_2$ Isothermal Vector Transport Data

The recent appearance of transference[33,54] and more extensive conductance data,[33] together with the diffusion data of this paper, now permit the calculation of l_{ij} for aqueous $MgCl_2$ from infinite dilution to saturation. A critical review and smoothing of the experimental data for t_1, Λ, and D_v follow. The activity data for $1 + m\, d \ln \gamma/dm$ have already been critically reviewed.[19]

A. t_1. The moving boundary (MB) data of Lee and Kay[54] to 0.5 mol·dm^{-3} and the emf data of Phang and Stokes[33] from 0.1 to 5.5 mol·kg^{-1} have been smoothed. The emf data of Schmidt[55] and MB data of Varshavskaya and Kaimakov[56] (at 20 °C) have been rejected as inaccurate and not a true measurement, respectively. The early MB measurements of Steele[57] and Denison[58,59] were not volume corrected. The early Hittorf measurements of Hittorf[60] at various temperatures between 6 and 15 °C are low; a later set[61] is within 0.01. A Hittorf point at 18 °C by Bein[62] is in good agreement. A set of Hittorf data by Drucker[63] at "room temperature" (about 18 °C) was claimed to have a minimum in t_{Cl} at 0.005 M, but the "minimum" is undoubtedly due to experimental scatter; these 18 °C data are in reasonably good agreement with the Lee–Kay MB results.

A preliminary reanalysis of the Phang–Stokes E_t vs. E data, with and without an a_0 term, gave the same values of t_1 to ±0.0002 for the fourth-order (E^4) polynomials

$$E_t = a_0 + a_1E + a_2E^2 + a_3E^3 + a_4E^4 \quad (3)$$

where E_t is their measured emf from concentration cells with transference and E is the value for a concentration cell without transference calculated from m and γ. An E^5 fit agreed with the E^4 fits to ±0.0001 in t_1 except at 0.1 m; consequently, an E^4 fit without a_0 was used. (a_0 should be zero in principle.)

The Phang–Stokes data were recalculated as follows. Their m, based on a molecular weight of 95.23 g·mol^{-1},[33] were converted to m based on our MW of 95.211 g·mol^{-1}. The E values were then obtained from our m and from γ calculated with the constants in columns two and three of Table VIII for eq 3 in ref 19, which contains critically reviewed osmotic-activity data. Their[33] $E(E_{ps})$ had been obtained from their m and from activity data taken from Robinson and Stokes.[64]

The recalculated E and the Phang–Stokes experimental E_t, when fitted with eq 3 up to E^4, yield $a_1 = 0.325\,968$, $a_2 = -0.420\,319$, $a_3 = 0.170\,663$, and $a_4 = 1.384\,506$, with a standard error $\sigma = 0.113$ mV. We obtained values of t_1 from the relation

$$t_1 = dE_t/dE \quad (4)$$

Sample comparisons between Phang–Stokes and our recalculated values show generally that $E - E_{ps}$ ranges from 0 to 1.7 mV (with the largest differences at high concentrations) and $t_1 - t_{1ps}$ ranges from +0.002 to −0.003 (percent errors of +0.6% at low concentrations to −2.0% at high concentrations).

It is hard to assess the accuracy of emf transference numbers, since the differentiation to get t_1 magnifies errors. Differences in t_1 between a third-order (E^3) fit and our chosen fourth-order fit cycle, in sign, are mostly 0.002–0.003, and as expected, are larger at the low (−0.009) and high (+0.010) concentration ends. Uncertainties are thus about ±0.005 in t_1 over most of the concentration range but close to 0.01 at the ends.

The Lee–Kay MB data and the recalculated emf t_1 at the molarities c corresponding to Phang–Stokes molalities m were plotted against $c^{1/2}$ in terms of the deviation function

$$w = t_1 - 0.1c - 0.3930 \quad (5)$$

As c decreases, the emf-based values become increasingly high compared to MB ones. A smooth curve can be drawn through the MB values from 0 to 0.4 M and through the emf values from 1 to 5 M. A smooth curve bridging the gap from 0.4 to 1 M was not more than 0.001 from the MB and was closer to emf points. Values of w were read off the curve at round values of c, and t_1 values were back-calculated.

In the dilute region, $0 \le c \le 0.02$ mol·dm^{-3}, the MB data were also plotted vs. c in terms of t_1' of the Lee–Kay equation[54]

$$t_1' = t_1 + 2.598(\tfrac{2}{3} - t_1)Bc^{1/2}/\Lambda^\circ \quad (6)$$

A smooth curve went through all the points to ±0.0002, and our intercept t_1° was 0.4112 ± 0.0004 compared to Lee–Kay's 0.4114 ± 0.0004. The values of t_1' depend on B (60.61) and on Λ° (129.67 cm^2·ohm^{-1}·equiv^{-1}), but a difference of 0.3 conductance units in Λ° changes t_1' by less than 0.0001. Values of t_1' were read off this curve at round values of c, and t_1 values were back-calculated. These t_1 values did not differ from those obtained from the w curve by more than ±0.0002.

We then calculated values of the emf t_1 at m corresponding to round values of c from 0.5 to 5 M using our fourth-order fit. These did not differ from those from the w curve by more than ±0.0003.

The final choices are the following: (a) $0 \le c \le 0.02$ mol·dm^{-3}. Values were taken from the t' graph (eq 6). (b) $0.02 < c \le 0.5$ mol·dm^{-3}. Values were taken from the w graph (eq 5). (c) $0.5 < c \le 5.0$ mol·dm^{-3}. Values were taken from the recalculated emf equation.

Estimated errors in these smoothed t_1 are (a) 0.0003–0.0004 from 0 to 0.4 M, based on the Lee–Kay estimate of 0.1%, (b)

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0.0005–0.001 from 0.4 to 0.5 M, and (c) 0.005 from 0.5 to 5.0 M, with perhaps larger errors to 0.01 near the high-concentration end, based on differences between third- and fourth-order emf fits.

B. Λ. We have used the conductance data of Shedlovsky and Brown,[23] Broadwater and Evans,[65] Fisher and Fox,[66] and Phang and Stokes.[33] The data of Phang–Stokes were recalculated to the old international ohm from their reported absolute ohm to preserve consistency with the extensive older tabulations of Λ. The following data have been rejected. Those of Van Rysselberghe et al.[67] are systematically high (probably due to the concentration errors noted by the authors); Postler's[68] data between 0.05 and 1.5 *M* are as much as 2 conductance units high; Suhrmann and Wiedersich's[69] values are about 1 unit high; Kondrat'ev and Nikich's values[70] are very low and scattered; those of Than and Amis[71] are mostly high; and Berecz and Bâder's data[72] are high and scattered at low concentrations and several tenths of a conductance unit low above 4 M.

The limiting conductance $\Lambda^\circ = 129.67$ was calculated from our value of $t_1^\circ = 0.4112$ and the standard value[64] for $\Lambda^\circ_{Cl} = 76.35$ cm²·ohm⁻¹·equiv⁻¹ from the equation

$$\Lambda^\circ = \Lambda^\circ_{Cl}/(1 - t_1^\circ) \quad (7)$$

The Broadwater and Evans' Λ° value[65] (130.0) is high, and Shedlovsky and Brown's value[23] (129.40) is low, whereas Lee and Kay's[54] (129.78, based on $t_1^\circ = 0.4114$ and $\Lambda^\circ_{Cl} = 76.39$ cm²·ohm⁻¹·equiv⁻¹) is in good agreement.

In the region $0 \le c \le 0.1$ mol·dm⁻³, the data of ref 23, 33, 65, and 66 were plotted vs. $c^{1/2}$ using the deviation function

$$w_1 = \Lambda + 129.4c^{1/2} - 129.4 \quad (8)$$

The data of Broadwater–Evans are systematically higher than those of Shedlovsky-Brown and of Fisher–Fox. Those of Phang–Stokes are also higher than those of Shedlovsky–Brown by about the same amount as Broadwater–Evans' data were, although there is no concentration overlap between those of Phang–Stokes and Broadwater–Evans. As mentioned by Lee and Kay, both Shedlovsky–Brown and Broadwater–Evans are noted for their care, so it is difficult to explain the discrepancy. (Contrary to the remark of Lee and Kay, Shedlovsky and Brown did use neutral $MgCl_2$ solutions). However, except for the six lowest concentration points, the Broadwater–Evans data are most consistent with the evaluation of Λ° by Lee–Kay and ourselves.

We have therefore subjectively drawn a smooth w_1 curve which starts at $c = 0$ with the w_1 corresponding to $\Lambda^\circ = 129.67$ cm²·ohm⁻¹·equiv⁻¹ and passes between the data of Shedlovsky–Brown and Broadwater–Evans but is closer to Broadwater–Evans up to $c = 0.001$ mol·dm⁻³. The first six Broadwater–Evans points are ignored, and the line is above and parallel to those of Shedlovsky–Brown. For c between 0.001 and 0.005 mol·dm⁻³, the line is above and parallel to Shedlovsky-Brown points and runs smoothly into the Phang–Stokes points. Between 0.005 and 0.1 mol·dm⁻³, the line runs through the Phang–Stokes points and is roughly parallel to those of Shedlovsky–Brown. The deviations of Broadwater–Evans and Phang–Stokes points from the line are mostly 0.03–0.05, with a few as large as 0.1 conductance unit.

Smoothed values of Λ were obtained by reading off w_1 from the curve for round values of c and back-calculating to get Λ. Overall, the uncertainties are ±0.05–0.1 conductance units.

For the region $0.1 < c \le 5$ mol·dm⁻³, we have relied on the Phang–Stokes data. A subjective line was drawn through their data by using the deviation function

$$w_2 = \Lambda + 48.25c^{1/2} - 110.00 \quad (9)$$

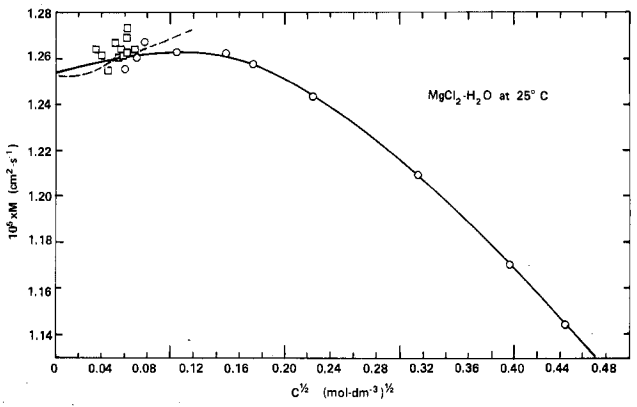


Figure 3. Thermodynamic diffusion coefficients M vs. $c^{1/2}$ for aqueous $MgCl_2$ at 25 °C in the dilute region: ○, this research, □, ref 6. The dotted line is the Onsager–Fuoss limiting law[73] for an ion size of 5.5 Å.

and w_2 plotted vs. $c^{1/2}$. Values of w_2 were read from the curve for round values of c and back-calculated to get Λ. Uncertainties are again estimated to be ±0.05–0.1 conductance units.

C. D_v. In the region $0 \le c \le 0.2$ mol·dm⁻³, we have relied on the data of Harned and Polestra[6] from 0.001 29 to 0.004 01 M and ours from 0.003 64 to 0.2 M. A convenient deviation function to smooth D_v in this region is the "thermodynamic diffusion coefficient" M of eq 2 (see Figure 3), with $(1 + m\ d\ \ln \gamma/dm)$ calculated from the same constants[19] mentioned in section IV-A above. Our D_v data are generally slightly lower ($\sim^3/_4$%) than the somewhat scattered points of Harned and Polestra, and ours have been emphasized in drawing the smooth M curve. Our three lowest concentration points deviate from it by at most 0.5%, and the rest are within 0.2% (except for one early measurement about 0.3% high). Values of M were read off for round values of c, and D_v values back-calculated using $1 + m\ d\ \ln \gamma/dm$ at the m corresponding to those round values of c. The value at $c = 0$, $D_v^\circ = 1.2538$ cm²·s⁻¹, was calculated from the Nernst–Hartley equation[13,64] using $\Lambda^\circ_{Cl} = 76.35$ and $\Lambda^\circ_{Mg} = 53.32$ cm²·ohm⁻¹·equiv⁻¹ (based on our $\Lambda^\circ = 129.67$ cm²·ohm⁻¹·equiv⁻¹).

In the region $0.2 < c \le 5$ mol·dm⁻³, our optical data were used. The less accurate data of ref 8 differ by many percent. The deviation function

$$z = D_v - [1.2538 - 0.935c^{1/2} + 1.182c - 0.415c^{3/2}] \quad (10)$$

was plotted vs. $c^{1/2}$, and a smooth curve drawn which emphasized our later, more accurate, base-line-corrected measurements. These later measurements are all within 0.2% of the line; the earliest set without base-line corrections are mostly as good with a maximum deviation of 0.5% except for one outlier (0.7%). Values of D_v were back-calculated from the z curve.

The curves from the M and z deviation functions were extended to overlap with each other. Agreement between the two is very good: 0.1% from 0.01 to 0.2 M, exact at 0.2 M, and less than 0.1% from 0.2 to 0.9 M. The overall uncertainties of 0.2–0.3% are not much greater than our experimental errors.

D. Comments on the Critically Reviewed Data. The smoothed values of t_1, Λ, D_v, and $1 + m\ d\ \ln \gamma/dm$ at 25 °C necessary to calculate the l_{ij} are given for round values of c in Table V along with the corresponding m and d. More figures than are signficant have been retained for consistency and reduction of accumulated round-off errors. The actual uncertainties in each quantity are given above. In general, graphical smoothing has been chosen over least-squares fitting, because an extensive analysis of equation fits showed that graphical values overlap better.

Note that theoretical equations were not used for diffusion or conductance at the lowest concentrations. $MgCl_2$ is an unsymmetrical electrolyte for which the theories are less accurate and ion sizes are uncertain. It is interesting, however, that the Onsager–Fuoss limiting law terms[73] for M with any plausible ion size indicate that as c increases, M initially declines, has a minimum

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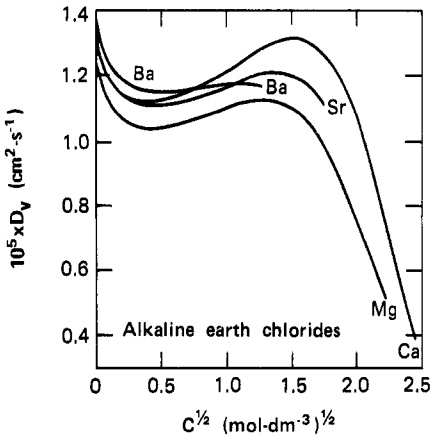


Figure 4. Diffusion coefficients D_v of alkaline-earth chlorides at 25 °C vs. $c^{1/2}$.

at 0.0001 M, and thereafter rises sharply (see dotted line in Figure 3). A change in slope as M approaches $M°$ at $c = 0$ seemed to be present on some unpublished plots of experimental data used to provide the smoothed results reported in ref 13; however, the data were too scattered to be certain, so no comment was made at that time.

V. Isothermal Ionic Transport Coefficients l_{ij}

The l_{ij} can be calculated from experimental data for special cases using the equation[13,15,20] for a binary system

$$\frac{l_{ij}}{N} = \frac{t_i t_j \Lambda}{10^3 F^2 z_i z_j} + \frac{r_i r_j D_v}{10^3 \hat{R} T r r_1 z_1 \left(1 + \frac{m \, \mathrm{d} \ln \gamma}{\mathrm{d}m}\right)} \quad (11)$$

where the l_{ij} are *solvent-fixed* transport coefficients, N is the normality in equiv·dm^{-3}, t_i is the transference number of ion i, z_i is its signed valence, r_i is its stoichiometric coefficient of ionization, F is the Faraday constant (96 493 C·equiv^{-1}), $\hat{R}$ is the gas constant (8.3144 J·mol^{-1}), T is the temperature in K (25 °C = 298.15 K), and $r = r_1 + r_2$. The l_{ij} coefficients have the units mol^2·J^{-1}·cm^{-1}·s^{-1}, l_{ij}/N the units mol^2·dm^3·J^{-1}·cm^{-1}·s^{-1}·equiv^{-1}, and Λ the units cm^2·ohm^{-1}·equiv^{-1}.

The smoothed data of Table V have been used in eq 11 to calculate l_{ij}/N in Table VI. Also included are the solvent molarity c_0 and the ionic coefficients R_{ij} for the inverse description of irreversible processes.[13] We have calculated the R_{ij} values from l_{ij} using eq 67, 68, 64, and 65 of ref 13. The l_{ij}/N and R_{ij} values have *not* been smoothed, and *more figures than are significant have been retained for consistency and to avoid round-off errors in future calculations*. From the estimated uncertainties of the smoothed data, the number of significant figures in l_{ij}/N and R_{ij} are at most three.

VI. Discussion

A. Diffusion Properties. The $MgCl_2$ D_v are lower than D_v for the other alkaline-earth chlorides[5,11,12] (Figure 4). As noted in more detail elsewhere,[12,16] the D_v values show no obvious regularities and have unexpected crossovers. However, the M data form a regular family of curves[12,16] in the order expected from conductivity considerations (Figure 5). This provides support for the irreversible thermodynamic concept that chemical potential gradients are the true driving forces for diffusion, not concentration gradients. The possible universality of these regularities in M will be tested on $MnCl_2$, $CdCl_2$, and $ZnCl_2$, now under investigation in this laboratory.

B. l_{ij} and the Onsager Reciprocal Relations. The l_{ij}/N values for $MgCl_2$ are shown in Figures 6–8, along with corresponding data for the 2–1 chlorides $CaCl_2$,[13,74] $BaCl_2$,[13] $CdCl_2$,[75] $ZnCl_2$,[76] and $NiCl_2$. The $NiCl_2$ values were calculated from the data of Stokes et al.[77] They and the $CdCl_2$ results are considered to be preliminary, owing to the uncertainties in D_v from diaphragm cells.

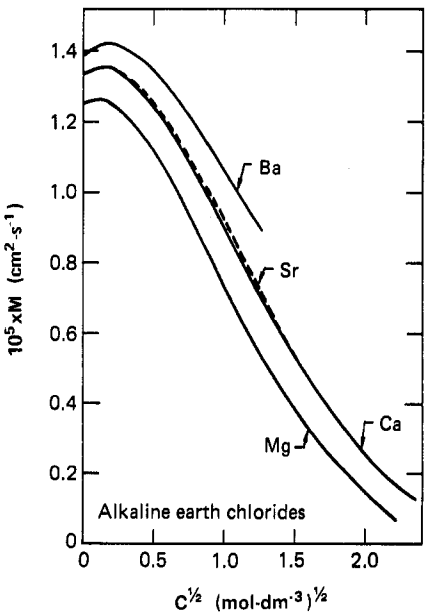


Figure 5. Thermodynamic diffusion coefficients M for alkaline-earth chlorides at 25 °C vs. $c^{1/2}$.

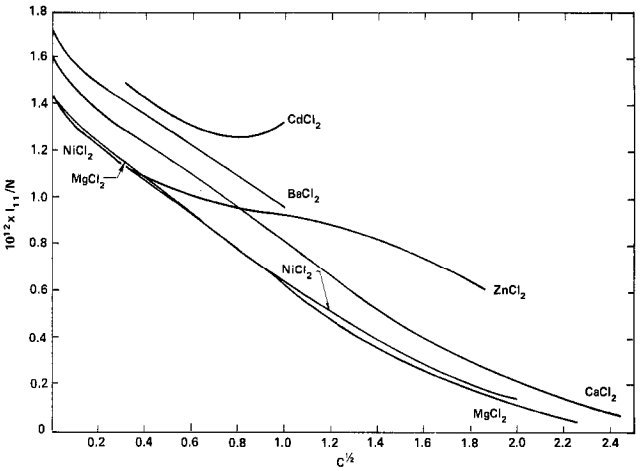


Figure 6. Cation intrinsic mobilities 10^{12} l_{11}/N of some alkaline-earth and transition-metal chlorides at 25 °C vs. $c^{1/2}$.

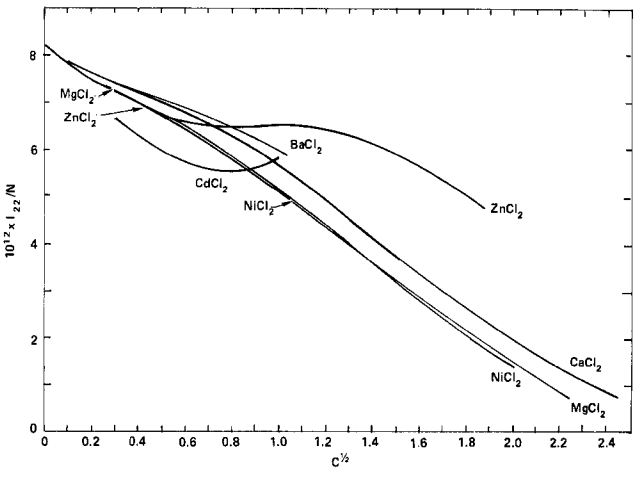


Figure 7. Anion intrinsic mobilities $10^{12} l_{22}/N$ of some alkaline-earth and transition-metal chlorides at 25 °C vs. $c^{1/2}$.

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TABLE V: Critically Reviewed Thermodynamic and Transport Data for Aqueous $MgCl_2$ at 25 °C[a-d]

c	m	t_1	Λ	$10^5 D_v$	$1 + m\,d\ln\gamma/dm$	d
0	0	0.4112	129.67	1.2538	1	0.997045
0.0001	0.0001_0	0.4081	127.13	1.2309	0.9807	0.99705
0.0005	0.0005_0	0.4047	124.20	1.2058	0.9594	0.99709
0.001	0.00100	0.4023	122.18	1.1892	0.9453	0.99713
0.002	0.0020_1	0.3991	119.45	1.1663	0.9276	0.99721
0.005	0.0050_2	0.3938	114.58	1.1349	0.8998	0.99745
0.01	0.0100_3	0.3892	110.13	1.1083	0.8777	0.99786
0.02	0.0200_7	0.3833	104.97	1.0835	0.8590	0.99866
0.05	0.0501_9	0.3725	97.17	1.0574	0.8497	1.00104
0.07	0.0702_9	0.3675	94.16	1.0497	0.8534	1.00261
0.1	0.1004_6	0.3612	90.74	1.0435	0.8633	1.00495
0.2	0.2012_8	0.3470	83.42	1.0379	0.9085	1.01267
0.3	0.3025_0	0.3372	78.58	1.0426	0.9631	1.02028
0.4	0.4041_2	0.3293	74.58	1.0506	1.0248	1.02788
0.5	0.5061_8	0.3221	71.19	1.0598	1.0923	1.03539
0.6	0.6086_9	0.3156	68.18	1.0682	1.1646	1.04285
0.7	0.7116_7	0.3095	65.44	1.0766	1.2412	1.05075
0.8	0.8151_4	0.3037	62.87	1.0843	1.3218	1.05760
0.9	0.9191_2	0.2983	60.49	1.0919	1.4061	1.06489
1.0	1.0236_3	0.2932	58.19	1.0992	1.4942	1.07212
1.5	1.5548_4	0.2696	48.13	1.1230	1.9907	1.10755
2.0	2.1020_2	0.2483	39.61	1.1181	2.5713	1.14189
2.5	2.66703	0.2282	32.24	1.0783	3.2006	1.17540
3.0	3.2513_9	0.2098	25.80	1.0023	3.8381	1.20831
3.5	3.85661	0.1939	20.20	0.8934	4.4762	1.24077
4.0	4.4845_6	0.1813	15.39	0.7635	5.1704	1.27279
4.5	5.1380_7	0.1729	11.35	0.6280	6.0439	1.30427
5.0	5.8215_6	(0.1702)	(7.86)	(0.4932)	7.2485	1.33493

[a] More places than are experimentally significant have been retained for consistency and to reduce the accumulation of errors in subsequent calculations. Sources of data and estimated errors of all quantities are given in the text. [b] Units are as follows: c, mol·dm^{-3}; m, mol·kg^{-1} of H_2O; Λ, cm^2·ohm^{-1}·equiv^{-1}; D_v, cm^2·s^{-1}; and d, g·cm^{-3}. [c] Extrapolated values (at c = 5.0 mol·dm^{-3}) are enclosed in parentheses. [d] Densities are calculated from molarity fit coefficients.

TABLE VI: Ionic Transport Coefficients for $MgCl_2$–H_2O at 25 °C[a-c]

c	c_0	$10^{12}l_{11}/N$	$10^{12}l_{12}/N$	$10^{12}l_{22}/N$	$10^{-11}NR_{11}$	$-10^{-11}N^{1/2}R_{12}$	$10^{-11}NR_{22}$	$-10^{-9}R_{10}$	$-10^{-9}R_{20}$	$10^{-8}R_{00}/N$
0	55.342	1.432	0	8.200	6.985		1.220	6.311	2.204	0.968
0.0001	55.342	1.412	0.039	8.159	7.081	2.369	1.226	6.337	2.185	0.967
0.0005	55.342	1.391	0.083	8.107	7.193	2.333	1.234	6.365	2.164	0.966
0.001	55.342	1.377	0.114	8.071	7.272	2.295	1.240	6.385	2.148	0.965
0.002	55.341	1.356	0.152	8.014	7.389	2.221	1.250	6.422	2.133	0.966
0.005	55.338	1.325	0.227	7.914	7.584	2.177	1.270	6.459	2.098	0.963
0.01	55.335	1.297	0.292	7.809	7.776	2.056	1.292	6.501	2.071	0.962
0.02	55.326	1.262	0.364	7.680	8.033	1.901	1.320	6.572	2.042	0.963
0.05	55.300	1.199	0.454	7.456	8.539	1.643	1.373	6.781	2.013	0.977
0.07	55.281	1.168	0.479	7.354	8.793	1.530	1.397	6.918	2.010	0.989
0.1	55.252	1.130	0.501	7.227	9.126	1.414	1.428	7.114	2.011	1.008
0.2	55.152	1.038	0.521	6.893	10.02	1.197	1.508	7.708	2.048	1.070
0.3	55.046	0.968	0.512	6.619	10.78	1.077	1.576	8.272	2.104	1.134
0.4	54.940	0.906	0.494	6.360	11.52	1.000	1.642	8.856	2.174	1.202
0.5	54.828	0.851	0.470	6.123	12.28	0.942	1.706	9.476	2.251	1.275
0.6	54.714	0.799	0.442	5.897	13.06	0.895	1.769	10.14	2.338	1.354
0.7	54.596	0.751	0.415	5.684	13.88	0.856	1.834	10.84	2.430	1.438
0.8	54.476	0.707	0.389	5.480	14.72	0.826	1.899	11.59	2.527	1.528
0.9	54.352	0.667	0.364	5.287	15.59	0.801	1.965	12.36	2.628	1.621
1.0	54.225	0.629	0.342	5.101	16.50	0.782	2.035	13.18	2.733	1.719
1.5	53.549	0.473	0.250	4.275	21.18	0.735	2.414	17.98	3.319	2.299
2.0	52.812	0.358	0.188	3.573	28.73	0.754	2.878	24.34	4.021	3.066
2.5	52.030	0.272	0.148	2.969	37.85	0.844	3.463	32.74	4.841	4.077
3.0	51.214	0.206	0.121	2.432	50.01	1.019	4.236	43.95	5.834	5.430
3.5	50.374	0.155	0.0988	1.946	66.86	1.283	5.310	59.62	7.171	7.342
4.0	49.509	0.113	0.0759	1.505	91.70	1.636	6.878	83.26	9.219	10.27
4.5	48.613	0.0790	0.0525	1.113	130.7	2.057	9.273	121.8	12.73	15.14
5.0	47.673	(0.0518)	(0.0318)	(0.764)	(198.0)	(2.608)	(13.43)	(190.4)	(19.52)	(24.06)

[a] Values in this table have been calculated directly from the smoothed data in Table V but have not been smoothed themselves. More places than are significant have been retained for consistency and for reduction in accumulation of errors in subsequent calculations. [b] Units are as follows: c, c_0, mol·dm^{-3}; N, equiv·dm^{-3}; l_{ij}/N, mol^2·dm^3·J^{-1}·cm^{-1}·s^{-1}·equiv^{-1}; and R_{ij}, J·cm·s·mol^{-2}. [c] Values based on extrapolations are enclosed in parentheses.

The $CaCl_2$ values of l_{ij}/N from ref 74 seem somewhat scattered between 1 and 3 M, and a smooth curve has been drawn through them.

The values of the intrinsic mobilities l_{11}/N and l_{22}/N of the cations and anions, respectively, generally decline with increasing concentrations, and at high enough concentrations fall to fairly low values for the strong electrolytes. Curves for the complexed electrolytes $ZnCl_2$ and $CdCl_2$ show unusual behavior including crossovers.

The values of the cation intrinsic mobilities l_{11}/N for the strong electrolytes are roughly parallel and in the order expected from the sizes of the hydrated ions and from limiting ionic conductances. On this basis, $NiCl_2$ is a strong electrolyte. The values for $ZnCl_2$ appear to be similar to $MgCl_2$ up to about 0.1 M; i.e., $ZnCl_2$

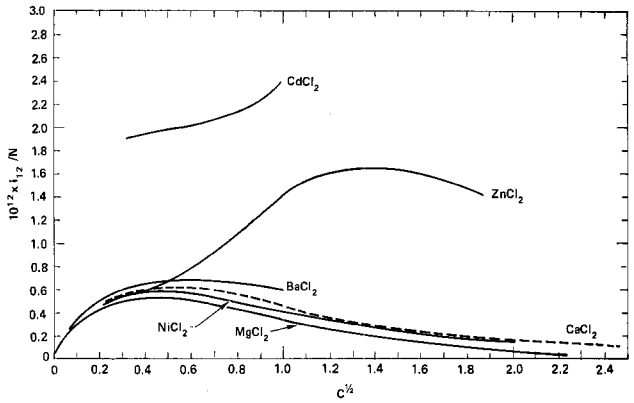


Figure 8. Interaction mobilities $10^{12}l_{12}/N$ of some alkaline-earth and transition-metal chlorides at 25 °C vs. $c^{1/2}$.

behaves as a strong electrolyte. Its values then increase relative to $MgCl_2$ and cross the $CaCl_2$ and extrapolated $BaCl_2$ curves. This behavior appears to be associated with ion pairing and complex ion formation, since at higher concentrations $ZnCl_2$ is known to be strongly self-complexed. The l_{11}/N values for $CdCl_2$ are puzzling. The limiting value at $c = 0$ for $CdCl_2$ should be nearly the same as for $MgCl_2$, $NiCl_2$, and $ZnCl_2$ because the limiting ionic conductances are nearly the same. This implies either a very strange behavior of the $CdCl_2$ l_{11}/N curve between 0 and 0.1 M or possible errors in the data used by McQuillan[75] to calculate l_{ij}/N. The D_v values for $CdCl_2$ have been remeasured in our laboratory over the whole concentration range and will be used later to resolve this question.

The l_{22}/N anion intrinsic mobilities approach the same limiting value at $c = 0$, as predicted theoretically.[13] As with l_{11}/N, the strong electrolytes are all similar in behavior, and the complexed electrolytes show unusual crossovers. The very small crossover for $NiCl_2$ and $MgCl_2$ is probably not significant, owing to uncertainties in the $NiCl_2$ data. $ZnCl_2$ appears to follow $MgCl_2$ to about 0.25 M, and then l_{22}/N diverges and crosses over the curves for the other salts. A separation between the chloride ion curves at finite concentrations is expected theoretically.[13,78,79] The order for the strong electrolytes is essentially the same as for the cation intrinsic mobilities and can be explained crudely as follows. In a binary system, ion motions are tied together by the electroneutrality condition. For example, in diffusion, a fast ion will be held back by a slow counterion and vice versa. In a comparison for a fixed anion, the overall relationships will thus follow the trends of the cations. Moreover, in a conductance experiment, a more hydrated slower cation (1) is more bulky, thus hindering anion movement, and (2) binds more water, leaving less solvent for the anions to move through. Both effects slow down the anions, and hence the anion movements also follow the trend of the cations. These examples indicate that intrinsic ionic mobilities are not as "intrinsic" as the name implies and as originally suggested.[13] Thus, cation–anion interactions contribute to intrinsic mobilities, although ion–solvent interactions and like-ion interactions are the largest part of l_{ii}/N.[78,79]

More illuminating are the low values of the $MgCl_2$ interaction coefficients l_{12}/N compared to those of other salts (Figure 8). This weaker cation–anion interaction implies a stronger electrolyte. This can be explained by a more strongly held, larger hydration sheath around the Mg^{2+} ion, compared to the case of Ba^{2+} for example, which would prevent the Cl^- ion from getting as close. Consquently, cation–anion interactions will not be as strong.

The size of l_{12}/N at any given concentration is seen to be larger for the weaker and more associated electrolytes. The same is true for 1–1 electrolytes.[13–15] This ordering corresponds generally to those obtained by comparison of complexation constants, association constants, or activity coefficients. *Thus, l_{12}/N serves as a sensitive characterizing parameter for the strength of an electrolyte within a given valence type.*

TABLE VII: Test of the ORR for Aqueous $MgCl_2$ at 25 °C

c, mol·dm^{-3}	t_1^h	t_1^c	δt_1^h	δt_1^c	l_{12}/l_{21}	δ
0.1	0.3612	0.3691	0.0004	0.009	1.077	0.091
0.2	0.3467	0.3511	0.0003	0.005	1.039	0.046
0.3	0.3369	0.3391	0.0003	0.005	1.018	0.044
0.4	0.3294	0.3301	0.0003	0.005	1.006	0.043
0.5	0.3229	0.3224	0.0005	0.005	0.996	0.043

On this basis, $MgCl_2$ is the strongest of these 2–1 chlorides. As noted previously[16] and above, $NiCl_2$ is a very strong electrolyte; it is stronger than $CaCl_2$ or $BaCl_2$, despite the fact that Ni is a transition metal. $CdCl_2$ is the most associated, and $ZnCl_2$ appears to act as a strong electrolyte to about 0.1 M and then rapidly becomes complexed.[16,76] This $ZnCl_2$ behavior is also indicated by l_{ii}/N and has recently been observed in certain special case coefficients such as tracer diffusion coefficients[80] and D_v.[81]

It is useful to review briefly some alternate but equivalent macroscopic formulations for describing linear transport processes. Our preferred choice is the *solvent-fixed* l_{ij} of irreversible thermodynamics,[13–16] which are reported here. The inverse coefficients R_{ij}, also reported here, have the apparent advantage of being independent of reference frame but do not seem to provide as direct an insight to electrolyte properties.[13,79] Another form of representing the inverse coefficients are the $\mathcal{D}_{ij}$ of Newman,[82] where

$$\mathcal{D}_{ij} = \frac{-\hat{R}T}{c_T R_{ij}} \tag{12}$$

and c_T is the sum of all ionic and solvent species concentrations. A fourth formulation uses the velocity correlation coefficients f_{ij} of Hertz and associates,[83–87] which are directly related to the *mass-fixed* Ω_{ij} of irreversible thermodynamics.[87–90] Since mass-fixed f_{ij} and Ω_{ij} can be obtained from solvent-fixed l_{ij}, and conversely,[88–90] the choice of formulation is a matter of taste or convenience.

Because the l_{ij} coefficients or their equivalents are more fundamental in principle than special case coefficients such as Λ or D_v, it was suggested[15] that electrolyte theorists devote their attention to calculation of the entropy production σ and the l_{ij} as begun by Onsager and Fuoss,[91] Onsager and Kim,[92] and Pikal.[79] We call attention to the recent work of Altenberger and Friedman[17] who obtained the l_{ij} by correlation function calculations, of Sändig[93,94] who used an extended Debye–Hückel formulation, and of Thacher and Lin[95] who also used correlation functions.

Finally, it is possible to test the Onsager reciprocal relations (ORR)[96]

$$l_{12} = l_{21} \tag{13}$$

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for aqueous $MgCl_2$. The values of l_{12}/l_{21} and the error δ of this ratio are obtained from eq 21 and 22 of ref 79. (Equation 21 of ref 79 should have F^2 instead of F.) These equations require t_1^h, the Hittorf (or moving boundary) transference number, obtainable from the MB data of Lee and Kay,[54] and t_1^c, the cell transference number, obtainable from the emf data of Phang and Stokes.[33] Unfortunately, these data do not have a very extensive overlap in concentration. Moreover, the emf data were not done to test the ORR, and thus there are not enough low-concentration points to properly tie down the derivative there. However, we have taken t_1^h from a "best" line through the Lee-Kay points on a w deviation plot (which differs slightly from the "best" line through the combined data) and t_1^c from the fourth-order $E_t(E)$ fit of section IV-A. The errors δt_1^h are as indicated by Lee and Kay, and δt_1^c are the differences between third- and fourth-order fits of $E_t(E)$. The Λ values were taken from Table V, and l_{12}/N from Table VI can be used because l_{12} and l_{21} are very close.

Examination of Table VII shows that the ORR are indeed satisfied, but unfortunately the error limits are relatively large. The verification of the ORR is somewhat better for aqueous $AgNO_3$,[79] $CdCl_2$,[75] $ZnCl_2$,[76] and $Zn(ClO_4)_2$[97] because the experiments were designed for that purpose and precision was emphasized.

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Registry No. $MgCl_2$, 7786-30-3; sucrose, 57-50-1.

Supplementary Material Available: A realignment procedure to optimize the use of a Beckman-Spinco Model H electrophoresis apparatus as a Rayleigh diffusiometer (7 pages). Ordering information is given on any current masthead page. Copies may also be obtained from the first two authors.

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Interpretation of the Silicon-29 Nuclear Magnetic Resonance Spectra of Zeolites: Synthetic Mazzite

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Interpretation of the ^{29}Si NMR spectrum of synthetic mazzite, which has a structure containing inequivalent crystallographic silicon atoms, has been made using existing information about the sensitivity of ^{29}Si chemical shifts to framework topology and gallium substitution. The assignment of the spectrum has been substantiated by comparison with elemental analysis data and by using a chemical model for the Si-Al distribution to calculate the spectral intensities.

Introduction

In cases where a zeolite structure contains only one crystallographically distinct tetrahedral framework silicon site, high-resolution, ^{29}Si NMR spectroscopy enables a quantitative determination of the relative populations of the five possible chemically distinct silicon environments (Si(nAl), $4 \geq n \geq 0$).[1] The integrated peak intensities measure the distributions of Si and Al atoms in the framework, which in turn enable inferences about pathways for zeolite crystallization to be drawn. From studies of the zeolites faujasite[2-6] and ZK4[7-9] over a wide composition range ($1.0 \leq Si/Al \leq 3.0$), it has been concluded that the distributions of Al atoms in the frameworks are always subject to the constraint of Loewenstein's rule,[10] which forbids the presence of Al-Al nearest neighbors. Superimposed on this is a second, weaker constraint that the number of Al-Al next-nearest neighbors is minimized. The latter has a dependence on framework topology since the Si-Al distributions in faujasite and ZK4 are different over the composition range.[9] The differences in observed distributions can be interpreted in terms of the pathways to the complete structures from isolated SiO_4 and AlO_4^- tetrahedra. It has been concluded that crystallization of the faujasite lattice probably occurs through a single 6-ring and that of ZK4 probably through a single 4-ring.[6] Important factors in selecting these systems for detailed treatments are the structural simplicity of the two zeolites, namely, the presence of single crystallographic silicon sites, and the broad range of composition accessible by direct synthesis. For most zeolites,

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